

Project Executable Files

Date	29 Jun 2026
Team ID	xxxxxx
Project Name	Smart Crop
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example similar) or	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes

10	Demo video or walkthrough (if required)	Yes
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Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

Smart Crop/

```

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|— static/          - CSS, JavaScript, Images
|— templates/       - HTML templates (Frontend pages)
|— venv/            - Python virtual environment (not uploaded to GitHub)
|— app.py           - Main Flask application
|— train_model.py    - Machine learning model training script
|— crop_model.pkl    - Trained machine learning model
|— Crop_recommendation.csv - Crop recommendation dataset
|— database.py       - Database-related operations
|— requirements.txt  - Project dependencies
|— README.md        - Project setup and execution guide
|— .gitignore       - Git ignore configuration
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://127.0.0.1:5000
Login Credentials (Demo)	nallagorlavarsha0@gmail.com Password : Varsha@2005
Platform / Hosting Provider	http://127.0.0.1:5000
Repository Link	https://github.com/Premalikhitha/SmartBridge-Project
Demo Video Link	https://drive.google.com/file/d/1O4oEjEwStissv-SboZu4ABrmYJ7oy3wv/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

- Pre-requisites
- ☐ Python 3.10 or above
 - ☐ Flask
 - ☐ NumPy
 - ☐ Pandas
 - ☐ Scikit-learn

- ☐ Jolbib
- ☐ Web browser (Chrome, Edge, or Firefox)
- ☐ Git (optional, for cloning the repository)

Step 1

Clone the repository:

```
git clone <GitHub Repository URL>
```

Step 2

Navigate to the project directory:

```
cd OptiCrop
```

Step 3

Install the required dependencies:

```
pip install -r requirements.txt
```

Step 4

Run the Flask application:

```
python app.py
```

Step 5

Open the URL displayed in the terminal (typically <http://127.0.0.1:5000/>) in your web browser.

Step 6

Enter the required soil and environmental parameters:

- ☐ Nitrogen (N)
- ☐ Phosphorous (P)
- ☐ Potassium (K)
- ☐ Temperature
- ☐ Humidity
- ☐ pH
- ☐ Rainfall

Click Predict to view the recommended crop.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	External atmospheric sensor weather polling delays due to public API quota throttling.	System automatically serves localized historical baseline parameter backups if API requests timeout.
2	Cold start delay (~30s) on free-tier hosting instance when system remains completely idle.	Fixed via ping script updates; allow an extra moment during initial evaluator login.